

PRad-II Target Change Instructions

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There are 4 target configurations relevant for standard production running:

- A.) ‘H₂ Production’: H₂ gas flows into the target cell, which then flows out of the cell windows and into the target chamber before being pumped away. The beamline overview screen will read “RG-O Target: H₂”.
- B.) ‘Empty Cell’: H₂ gas flows into the target chamber but not into the target cell. The beamline overview screen will read “RG-O Target: Empty Cell”.
- C.) ‘Empty Cell & Chamber’: Target cell is in the beamline but there is no gas in either the cell or chamber. The beamline overview screen will read “RG-O Target: Empty Cell & Chamber”.
- D.) ‘No Targets/Out’: The target ladder (containing the cell and aluminum/carbon sheets) is completely removed from the beamline. The beamline overview status will read “Out”.

Below is the PRad-II Target overview screen under nominal full-target H₂ operating conditions.

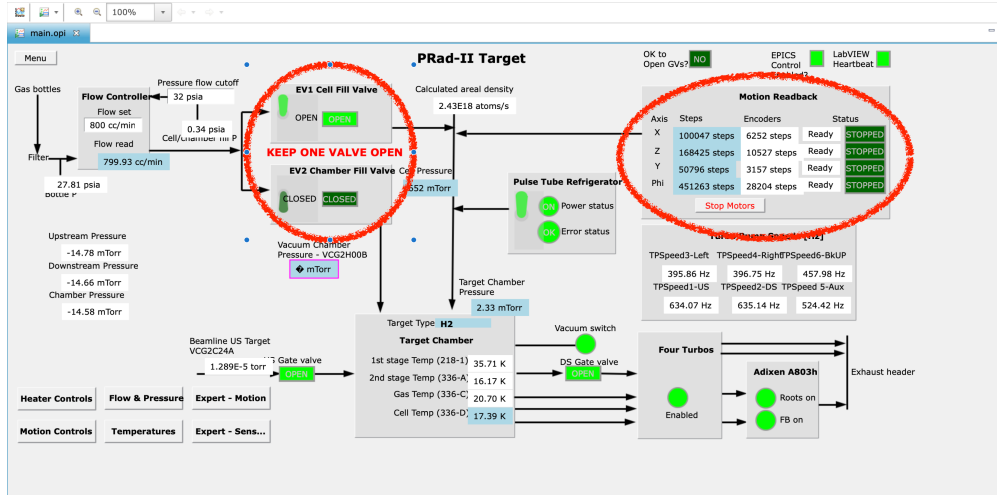


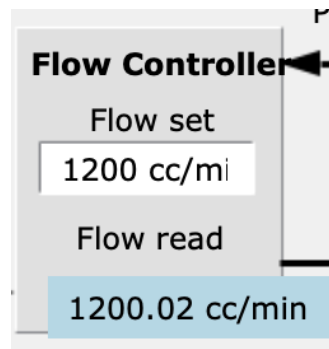
Figure 1: PRad-II Target H₂ Overview. NOTE TO SHIFT TAKERS: All relevant information for the checklist or run tracking have their text fields highlighted in blue.

The current status of the target is calculated from the position of the EV1/EV2 switches and the Y motor readback, which are circled in red in Fig 1. **Only EV1 or EV2 can be open at one time.** Both can be closed at the same time.

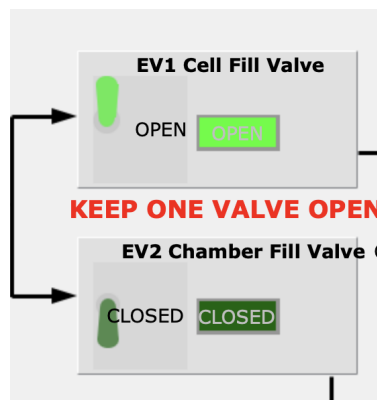
- A. When EV1 is open, and $Y = 72490$ the target status is H_2 .
- B. When EV2 is open, and $Y = 72490$, the status is Empty Cell.
- C. When EV1/EV2 are both closed, and $Y = 72490$, the status is “Empty Cell & Chamber”.
- D. When EV1/EV2 are closed, and $Y = 7300000$, the target is out and the status is “Out”.
- E. When EV1/EV2 are closed, and $Y = 5789167$, the carbon foil is in and the target is “C”.
- F. When EV1/EV2 are closed, and $Y = 4366935$, the aluminum foil is in and the target is “Al”.

1 Changing Target Configurations

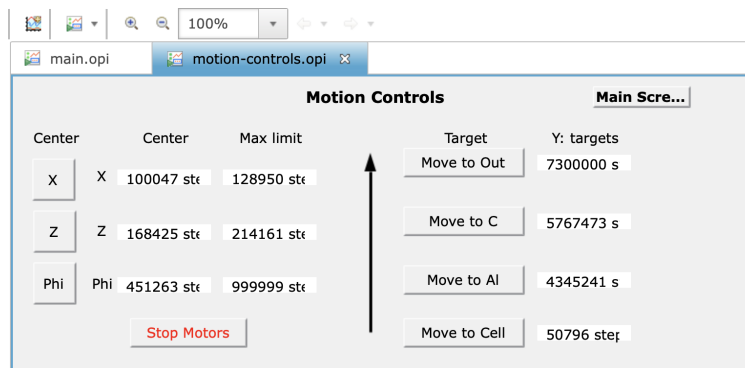
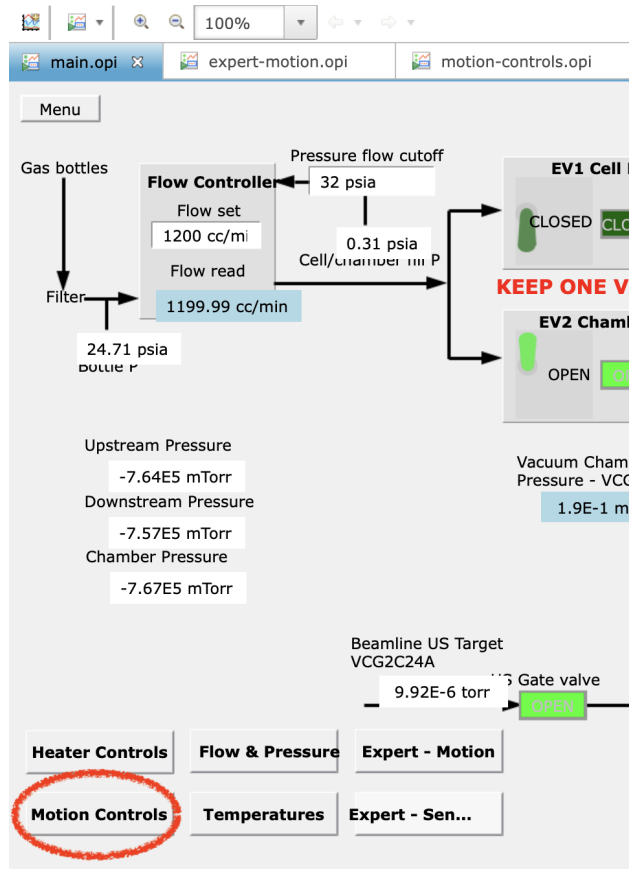
1. **Stop the current run.**
2. **Flow set must be 0 when changing target configurations.** In the 'Flow Controller' field located in the upper left corner, put the 'Flow set' to 0. Flow must be set to 0 when changing configurations. The flow read may still be around 10 cc/min when flow is set to 0. This is due to the minimum sensitivity of the flow controller and not a cause for concern.



3. **Once flow is set to zero, now ensure that EV1 and EV2 are both closed.** You may close them by clicking on the toggle switch.



4. If your target configuration change does not require motion of the target cell, skip to step 6. If you are using the motors to move the target cell position, **call MCC and have them take away the beam and insert the dumplette.** Turn off the GEM HVs. Once that is complete you are ready to open the “Motion Controls” panel on the GUI.



- On this screen, you can hit one of the buttons to move the target ladder motors to one of the 4 preset Y positions. 'Cell' is for nominal production running. 'Out' is to remove the target ladder from the target chamber. Configurations A-C use "Cell". Only configuration D uses "Out". Once you have hit one of these buttons, return to the main screen. You should now see in the upper right the motor status for Y change from "STOPPED" to "MOVING".

Motion Readback				
Axis	Steps	Encoders	Status	
X	100047 steps	6252 steps	Ready	STOPPED
Z	168425 steps	10527 steps	Ready	STOPPED
Y	50796 steps	3157 steps	Ready	STOPPED
Phi	451263 steps	28204 steps	Ready	STOPPED

Stop Motors

6. Once the target is in the correct position for your desired configuration, the motor status will once again change to “STOPPED.” If you followed the instructions up until now, no matter what your desired end result, the status of the target *should* be:

- flow set to 0
- both EV1/EV2 closed
- motor readbacks reading STOPPED.
- Target Type listed above the target chamber temperatures shows one of the following statuses: “Empty Cell & Chamber”, “Out”, “C”, or “Al”.

If you are running:

- **‘Configuration A’=‘Full Target’=‘H₂’:**
open EV1 and then set flow to 1200 cc/m. If you had to move the target to get to this configuration, turn on the GEM HVs, call MCC to remove the dumplette and to restore beam. When the beam is stable at the requested current and GEM HVs are on, prestart then start the run.
- **‘Configuration B’=‘Empty Cell’:**
open EV2 and then set flow 1200 cc/m. If you had to move the target to get to this configuration, turn on the GEM HVs, call MCC to remove the dumplette and to restore beam. When the beam is stable at the requested current and GEM HVs are on, prestart then start the run.
- **‘Configuration C’=‘Empty Cell & Chamber’:**
If you had to move the target to get to this configuration, turn on the GEM HVs, call MCC to remove the dumplette and to restore beam. When the beam is stable at the requested current and GEM HVs are on, prestart then start the run.
- **‘Configuration D’ = OUT, or Carbon or Aluminum:**
turn on the GEM HVs, call MCC to remove the dumplette and to restore beam. When the beam is stable at the requested current and GEM HVs are on, prestart then start the run.

Motion Readback					
Axis	Steps	Encoders	Differential	Status	
X	95813 steps	5990 steps	2 steps	Ready	STOPPED
Z	168425 steps	10526 steps	1 steps	Ready	STOPPED
Y	72490 steps	4531 steps	0 steps	Ready	STOPPED
Phi	228840 steps	14303 steps	1 steps	Ready	STOPPED
Stop Motors					

Figure 2: The Differential column measures the number of steps lost between the true encoder value. If after moving from 'Out' to 'Cell' this "Differential" value ever exceeds 100, you should "home" the motor and attempt to move it to the nominal Y position again.

2 Lost Motor Steps

The target is positioned with stepper motors. If the motor loses steps, the readback value of the motor position will differ from the true encoder value. This can be seen in the "Differential" column of the motor status readback. When moving the target from 'Cell' to 'Out', the motor may lose a significant amount of steps. The accumulation of missing steps is not very important for moving to 'Out', but when moving back to 'Cell' the accumulated lost steps can become problematic. I would recommend that if on your trip back from 'Out' to 'Cell' the "Differential" value ever exceeds 100, you should "home" the Y motor using the 'Expert - Motion' tab on the target GUI. Only hit the "home" button for the Y motor. After doing so, the 'Go to Cell' button will no longer work, since you have already clicked it once. Once the motor has been homed, the step and encoder position will say 0. Return to the 'Expert - Motion' screen and enter the Y position for 'Cell' 72490. The number may already be in that field, in which case, delete the last digit, retype it, and then hit enter. You should be prompted with a pop-up asking if you are sure you want to move the motor. Click "Yes". This should move the target back to the nominal Y position for 'Cell'. Repeat this process of homing the motor and moving to 'Cell' until the "Differential" value is less than 20. If it takes more than 4 tries, call Andrew.